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Sorting

Q.No1:

Problem Description:

Tina owns a match making company, which even to her surprise is an extreme hit. She says that her success rate cannot be matched (Yes, letterplay!) in the entire match-making industry. She follows an extremely simple algorithm to determine if two people are matches for each other. Her algorithm is not at all complex, and makes no sense - not even to her. But she uses it anyway.

Let's say that on a given day she decides to select n people - that is, n boys and n girls. She gets the list of n boys and n girls in a random order initially. Then, she arranges the list of girls in ascending order on the basis of their height and boys in descending order of their heights. A girl A_i can be matched to a boy on the same index only, that is, B_i and no one else. Likewise, a girl standing on A_k can be only matched to a boy on the same index B_k and no one else.

Now to determine if the pair would make an ideal pair, she checks if the modulo of their heights is 0, i.e., $A_i \% B_i == 0$ or $B_i \% A_i == 0$. Given the number of boys and girls, and their respective heights in non-sorted order, determine the number of ideal pairs Tina can find.

Constraints:

$1 \leq \text{Test Cases} \leq 10^2$

$1 \leq N \leq 10^4$

$1 \leq A_i, B_i \leq 10^5$

Input Format:

The first line contains number of test cases. Then, the next line contains an integer, n , saying the number of boys and girls. The next line contains the height of girls, followed by the height of boys.

Output Format:

Print the number of ideal pairs in a separate lines

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
2
4
1 6 9 12
4 12 3 9
4
2 2 2 2
2 2 2 2
```

EXPECTED OUTPUT

```
2
4
```

Test Case 2

INPUT (STDIN)

```
2
4
3 6 9 12
4 12 16 9
4
1 1 1 1
2 2 2 2
```

EXPECTED OUTPUT

```
3
4
```

Code:

```
#include <stdio.h>

void sortGirls(int a[], int n)
{
    int i, j, temp;

    for (i = 0; i < n; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (a[i] > a[j])
            {
                temp = a[i];
                a[i] = a[j];
                a[j] = temp;
            }
        }
    }
}

void sortBoys(int a[], int n)
{
    int i, j, temp;

    for (i = 0; i < n; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (a[i] < a[j])
            {
                temp = a[i];
                a[i] = a[j];
                a[j] = temp;
            }
        }
    }
}

int main()
{
    int T;
    scanf("%d", &T);

    int answer[100];

    for (int t = 0; t < T; t++)
```

```

{
    int n;
    int girls[10000];
    int boys[10000];

    scanf("%d", &n);

    for (int i = 0; i < n; i++)
    {
        scanf("%d", &girls[i]);
    }

    for (int i = 0; i < n; i++)
    {
        scanf("%d", &boys[i]);
    }

    sortGirls(girls, n);
    sortBoys(boys, n);

    int count = 0;

    for (int i = 0; i < n; i++)
    {
        if (girls[i] % boys[i] == 0 ||
            boys[i] % girls[i] == 0)
        {
            count++;
        }
    }

    answer[t] = count;
}

printf("output: \n");
for (int t = 0; t < T; t++)
{
    printf("%d\n", answer[t]);
}

return 0;
}

```

Output:

Test_case 1:

```
addityajaiswal@Addityas-MacBook-Air:~$ cat array.txt
2
4
1 6 9 12
4 12 3 9
4
2 2 2 2
2 2 2 2
output:
2
4
```

Test_case 2:

```
addityajaiswal@Addityas-MacBook-Air:~$ cat array.txt
2
4
3 6 9 12
4 12 16 9
4
1 1 1 1
2 2 2 2
output:
3
4
```